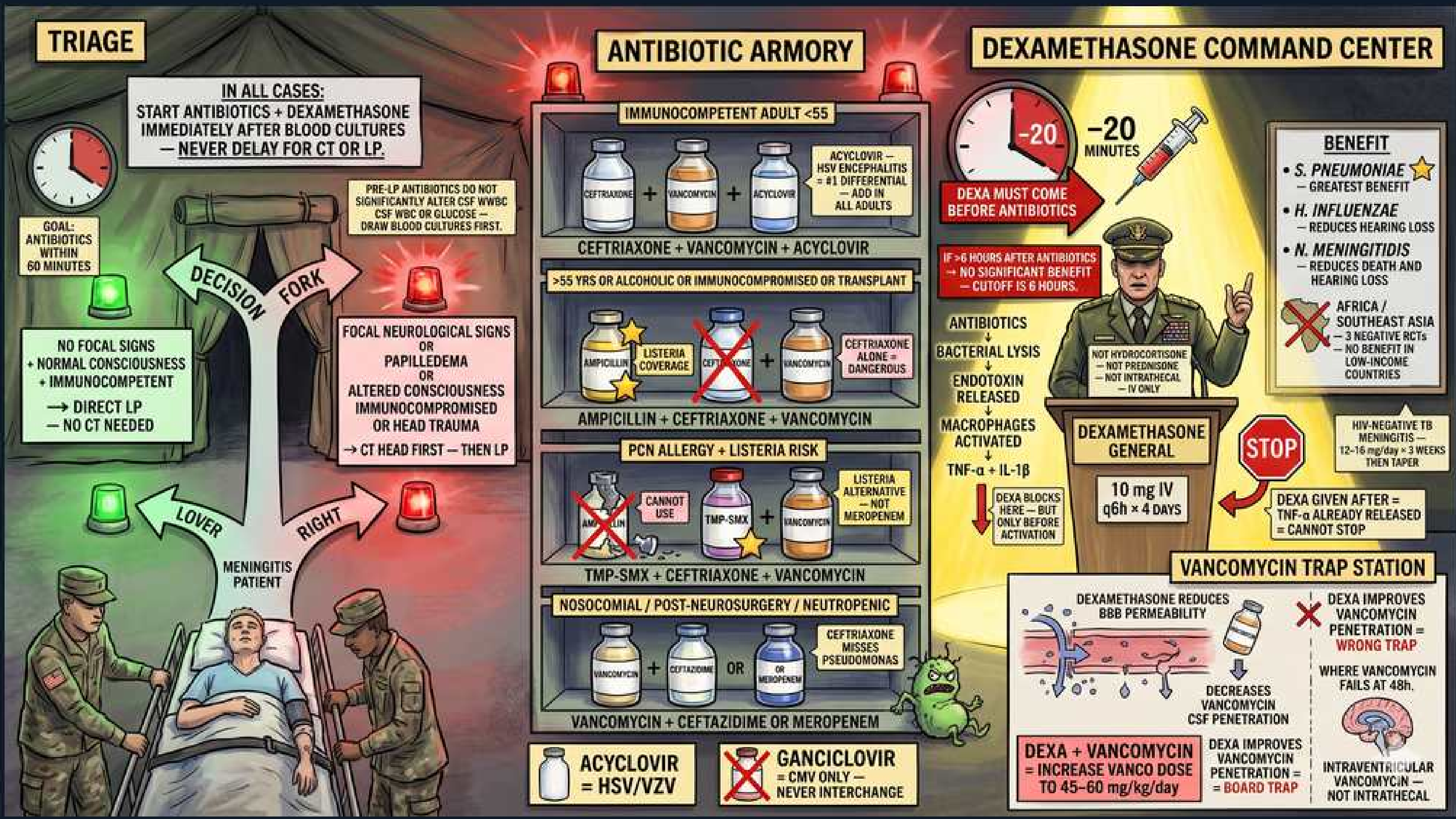
Bacterial Meningitis — Empiric Antibiotics, Dexamethasone & Timing



1. Antibiotics before CT/LP

WHAT YOU SEE

Triage rule: start antibiotics + dexamethasone after cultures, never delay for CT/LP

WHAT IT ENCODES

In all suspected bacterial meningitis, draw blood cultures then give empiric antibiotics + dexamethasone immediately. Never delay antibiotics waiting for CT or LP — goal is treatment within 60 minutes.

BOARD TRAP

Wrong: always image and tap before any antibiotic. Delaying antibiotics for imaging increases mortality.

---

2. When to CT before LP

WHAT YOU SEE

Decision fork: focal signs/papilledema/immunocompromised/altered → CT first

WHAT IT ENCODES

CT head before LP is indicated only with focal neuro deficits, papilledema, new seizures, altered consciousness, immunocompromise, or history of CNS disease. Otherwise tap directly. Antibiotics go in BEFORE the CT regardless.

BOARD TRAP

Wrong: every meningitis patient needs CT before LP. Low-risk patients can be tapped directly — and antibiotics never wait for CT.

---

3. Ceftriaxone + vancomycin (immunocompetent)

WHAT YOU SEE

Armory: immunocompetent adult <55 = ceftriaxone + vancomycin

WHAT IT ENCODES

Empiric regimen for healthy adults <50–55: vancomycin (for resistant S. pneumoniae) + ceftriaxone (covers pneumococcus and meningococcus). Vancomycin is added because of cephalosporin-resistant pneumococcus.

BOARD TRAP

Ceftriaxone alone is wrong — you must add vancomycin for resistant pneumococcus.

---

4. Add ampicillin for Listeria

WHAT YOU SEE

Age >50 or alcoholic/immunocompromised → add ampicillin for Listeria coverage

WHAT IT ENCODES

Patients >50, pregnant, alcoholic, or immunocompromised need ampicillin added to vanc + ceftriaxone to cover Listeria monocytogenes. Cephalosporins do NOT cover Listeria.

BOARD TRAP

Forgetting ampicillin in the elderly/immunocompromised — ceftriaxone misses Listeria.

---

5. PCN allergy → TMP-SMX

WHAT YOU SEE

PCN allergy with Listeria risk = TMP-SMX replaces ampicillin

WHAT IT ENCODES

In penicillin-allergic patients who still need Listeria coverage, substitute TMP-SMX (or meropenem) for ampicillin. Meropenem also covers gram-negatives and Listeria.

BOARD TRAP

Dropping Listeria coverage entirely in a PCN-allergic elderly patient — use TMP-SMX/meropenem instead.

---

6. Nosocomial/post-neurosurgery

WHAT YOU SEE

Vancomycin + cefepime or meropenem for nosocomial/neutropenic

WHAT IT ENCODES

Healthcare-associated meningitis (post-neurosurgery, shunt, penetrating trauma, neutropenic) needs antipseudomonal coverage: vancomycin plus cefepime, ceftazidime, or meropenem. Standard ceftriaxone misses Pseudomonas.

BOARD TRAP

Ceftriaxone misses Pseudomonas — use cefepime/ceftazidime/meropenem in nosocomial cases.

---

7. Acyclovir for HSV/VZV

WHAT YOU SEE

Add acyclovir if HSV/VZV encephalitis suspected

WHAT IT ENCODES

If encephalitis (altered mentation, focal/temporal signs, seizures) is possible, add IV acyclovir empirically for HSV/VZV while awaiting CSF PCR — do not wait.

BOARD TRAP

Ganciclovir is for CMV, not the empiric HSV/VZV add-on; acyclovir is the encephalitis empiric.

---

8. Dexamethasone before/with antibiotics

WHAT YOU SEE

Command center: dexamethasone must come before antibiotics, within 20 min

WHAT IT ENCODES

Give dexamethasone (10 mg IV q6h x4 days) with or just before the first antibiotic dose. It blunts the inflammatory surge from bacterial lysis. If given after antibiotics, the cytokine cascade is already released and benefit is lost.

BOARD TRAP

Giving dexamethasone after antibiotics = no benefit. It must precede or accompany the first dose.

---

9. Dexamethasone benefits S. pneumoniae

WHAT YOU SEE

Benefit list: greatest in S. pneumoniae, also H. influenzae

WHAT IT ENCODES

Dexamethasone clearly reduces mortality and hearing loss in pneumococcal meningitis and reduces deafness in H. influenzae. Continue only if cultures confirm pneumococcus; stop otherwise.

BOARD TRAP

Steroid benefit is strongest for S. pneumoniae — stop dexamethasone if a different organism is found.

---

10. Stop dexa if not pneumococcus

WHAT YOU SEE

Stop sign: discontinue if organism is not pneumococcus

WHAT IT ENCODES

Once CSF/culture shows a non-pneumococcal organism (e.g., meningococcus, Listeria), discontinue dexamethasone — benefit is unproven and it may impair antibiotic CNS penetration.

BOARD TRAP

Continuing steroids regardless of organism — stop once non-pneumococcal cause is confirmed.

---

11. Dexa lowers vancomycin CNS penetration

WHAT YOU SEE

Vancomycin trap station: dexa reduces BBB permeability

WHAT IT ENCODES

Dexamethasone reduces blood-brain-barrier permeability and thus lowers vancomycin CSF levels. To compensate, dose vancomycin higher (target trough 15–20) and confirm clinical response.

BOARD TRAP

Board point: with dexamethasone, vancomycin CNS levels drop — increase the vanc dose, do not switch to intrathecal.

---

12. No intrathecal vancomycin routinely

WHAT YOU SEE

Warning: vancomycin IV, not intrathecal in standard meningitis

WHAT IT ENCODES

Standard bacterial meningitis is treated with IV vancomycin (higher dose), not intrathecal. Intrathecal/intraventricular vanc is reserved for refractory shunt/ventriculitis cases.

BOARD TRAP

Choosing intrathecal vancomycin for routine meningitis — give higher-dose IV instead.

---

13. Listeria patient cue

WHAT YOU SEE

Patient with green Listeria mascot in triage

WHAT IT ENCODES

The green character flags Listeria risk groups (elderly, pregnant, immunocompromised, alcoholic) — the trigger to add ampicillin to empiric therapy.

BOARD TRAP

Listeria is the organism cephalosporins miss; recognize the at-risk patient to add ampicillin.

---

14. 60-minute door-to-antibiotic goal

WHAT YOU SEE

Goal: antibiotics within 60 minutes

WHAT IT ENCODES

Time to effective antibiotics drives outcome — target administration within 60 minutes of presentation. Blood cultures first, then immediate empiric therapy plus dexamethasone.

BOARD TRAP

Any workflow that pushes antibiotics past an hour for imaging or LP worsens mortality.

---

15. Epidemiology of dexa benefit

WHAT YOU SEE

Region note: pneumococcal-predominant settings benefit most

WHAT IT ENCODES

Adjunctive dexamethasone shows the clearest mortality benefit in high-income, pneumococcal-predominant settings; benefit is less established where TB/HIV meningitis predominate.

BOARD TRAP

Assuming universal steroid benefit — context (organism, region) determines net benefit.
